

SecureCode-FL: A Federated Learning Framework for Real-Time Code Vulnerability Detection in Integrated Development Environments

Research Project Report

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Abstract

This report presents SecureCode-FL, a novel federated learning framework designed to detect code vulnerabilities in real-time within integrated development environments. The system addresses critical challenges in software security by combining machine learning-based vulnerability detection with privacy-preserving federated learning techniques. Unlike traditional centralized approaches, SecureCode-FL ensures that sensitive source code never leaves organizational boundaries, making it particularly suitable for enterprises subject to strict data governance regulations such as GDPR, HIPAA, and industry-specific compliance requirements. We propose a practical implementation as a Visual Studio Code extension, discuss the theoretical foundations, review relevant literature, analyze the business case for enterprise adoption, and outline a feasible development roadmap. Our framework demonstrates how organizations can collaboratively improve vulnerability detection models while maintaining complete data sovereignty.

1 Introduction

Software vulnerabilities represent one of the most significant cybersecurity challenges facing modern organizations. The 2024 State of Software Security report indicates that 76% of applications contain at least one security flaw, with the average application containing 26 vulnerabilities [1]. Traditional static analysis tools, while useful, suffer from high false-positive rates and inability to adapt to emerging vulnerability patterns. Meanwhile, machine learning approaches to vulnerability detection have shown promise but typically require centralized data collection, creating insurmountable barriers for organizations with strict data privacy requirements.

Federated Learning (FL), introduced by McMahan et al. [2], offers a paradigm shift: enabling collaborative machine learning without centralizing sensitive data. This approach is particularly well-suited to code vulnerability detection, where source code represents intellectual property that organizations are legally and strategically compelled to protect.

1.1 Problem Statement

Current code vulnerability detection systems face three primary challenges:

1. **Privacy Concerns:** Centralized ML systems require uploading source code to external servers, violating data sovereignty and compliance requirements.
2. **Real-time Detection Gap:** Most vulnerability scanners operate as batch processes, failing to provide immediate feedback during development.
3. **Adaptation Limitations:** Static rule-based systems cannot learn from new vulnerability patterns without manual updates.

1.2 Proposed Solution

SecureCode-FL addresses these challenges through:

- A federated learning architecture that trains vulnerability detection models collaboratively across organizations without data sharing
- Real-time inference within the IDE, providing immediate feedback to developers
- Continuous model improvement through privacy-preserving aggregation of local model updates
- A practical VS Code extension implementation with minimal performance overhead

2 Literature Review

2.1 Machine Learning for Vulnerability Detection

The application of machine learning to vulnerability detection has evolved significantly over the past decade. Early work by Yamaguchi et al. [3] demonstrated that graph-based code representations could effectively identify vulnerability patterns. More recently, deep learning approaches have shown superior performance.

Deep Learning Approaches: Li et al. [4] introduced VulDeePecker, using bidirectional LSTM networks on code gadgets to detect vulnerabilities with 92.1% accuracy. Russell et al. [5] extended this work with attention mechanisms, improving detection of memory-related vulnerabilities. Zhou et al. [6] proposed Devign, a graph neural network approach that captures both syntactic and semantic code properties, achieving F1-scores of 0.63 on real-world datasets.

Transformer-Based Models: The introduction of large language models has revolutionized code analysis. CodeBERT [7], trained on 6.4 million code-documentation pairs, demonstrated strong performance on vulnerability detection when fine-tuned. More recently, Hanif and Maffeis [8] showed that domain-specific pre-training on vulnerability datasets significantly improves detection accuracy, achieving 96.3% F1-score on the SARD dataset.

Limitations: Despite impressive results, these approaches require centralized training on large codebases, making them impractical for privacy-sensitive organizations.

2.2 Federated Learning Fundamentals

Federated Learning, formalized by McMahan et al. [2], enables collaborative model training across decentralized devices. The FederatedAveraging algorithm aggregates local model updates:

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1} \quad (1)$$

where w_{t+1} is the global model at round $t + 1$, w_k^{t+1} are local models from K clients, and n_k/n represents the proportion of data held by client k .

Privacy Enhancements: Differential privacy mechanisms [9] can be integrated to provide formal privacy guarantees. The addition of calibrated noise to gradients ensures that individual training examples cannot be reconstructed:

$$\tilde{g}_t = g_t + \mathcal{N}(0, \sigma^2 C^2 I) \quad (2)$$

where C is the clipping threshold and σ controls the privacy-utility tradeoff.

Secure Aggregation: Bonawitz et al. [10] introduced secure aggregation protocols using cryptographic techniques, ensuring that the central server cannot observe individual client updates, only their aggregate.

2.3 Federated Learning in Security Applications

Recent work has explored FL in cybersecurity contexts. Nguyen et al. [11] applied FL to intrusion detection systems, demonstrating that federated models achieve comparable accuracy to centralized approaches while preserving privacy. Zhao et al. [12] showed that federated learning can effectively detect Android malware across organizations without sharing app data.

Code-Specific Applications: Pasquini et al. [13] explored federated learning for code authorship attribution, demonstrating feasibility in code analysis tasks. However, to our knowledge, no prior work has specifically addressed federated vulnerability detection with real-time IDE integration.

2.4 IDE-Based Security Tools

Modern IDEs increasingly incorporate security features. GitHub Copilot and Amazon CodeWhisperer provide AI-assisted coding but operate on cloud infrastructure, raising privacy concerns. IntelliCode [14] uses local models for suggestions but lacks vulnerability-specific training. SpotBugs and SonarLint provide static analysis but cannot adapt through learning.

3 Methodology

3.1 System Architecture

SecureCode-FL employs a three-tier architecture:

1. **Client Layer:** VS Code extension running locally, performing inference and local training

2. **Aggregation Layer:** Federated server coordinating global model updates
3. **Model Layer:** Neural network architecture optimized for vulnerability detection

3.1.1 Client-Side Component

The VS Code extension operates as a Language Server Protocol (LSP) implementation, providing:

- Real-time code analysis on file save or configurable intervals
- Inline diagnostics highlighting potential vulnerabilities
- Code action suggestions for remediation
- Background model training on local code repositories

3.1.2 Model Architecture

We propose a hybrid architecture combining:

Code Representation: Graph Neural Networks (GNNs) process abstract syntax trees (ASTs) and program dependence graphs (PDGs) to capture structural code properties:

$$h_v^{(l+1)} = \sigma \left(W^{(l)} h_v^{(l)} + \sum_{u \in \mathcal{N}(v)} \frac{1}{|\mathcal{N}(v)|} h_u^{(l)} \right) \quad (3)$$

Sequence Encoding: A lightweight transformer encoder processes token sequences for semantic understanding.

Classification Head: Multi-label classification identifies vulnerability types (e.g., SQL injection, XSS, buffer overflow, path traversal).

3.1.3 Federated Training Protocol

The training process follows a modified FedAvg algorithm:

3.2 Data Preparation

3.2.1 Training Data Sources

Organizations can leverage multiple data sources:

- Internal code repositories with known vulnerabilities from past security audits
- Synthetic vulnerable code generated through mutation testing
- Public datasets (CVE database, NVD) for initial model training

Algorithm 1 SecureCode-FL Training Protocol

Server:

Initialize global model w_0
for each round $t = 1, 2, \dots$ **do**
 Select subset S_t of clients
 for each client $k \in S_t$ **in parallel do**
 $w_k^{t+1} \leftarrow \text{ClientUpdate}(k, w_t)$
 end for
 $w_{t+1} \leftarrow \sum_{k \in S_t} \frac{n_k}{n} w_k^{t+1}$
 Apply differential privacy noise (optional)
end for

ClientUpdate(k, w):

$\mathcal{B} \leftarrow$ split local data into batches
for each local epoch $i = 1$ to E **do**
 for batch $b \in \mathcal{B}$ **do**
 $w \leftarrow w - \eta \nabla \ell(w; b)$
 end for
end for
return w to server

3.2.2 Code Preprocessing Pipeline

1. **Parsing:** Convert source code to AST using tree-sitter or similar parsers
2. **Normalization:** Abstract variable names, literal values to reduce dimensionality
3. **Slicing:** Extract code slices relevant to potential vulnerabilities
4. **Feature Extraction:** Generate graph and sequence representations

3.3 Privacy and Security Measures

3.3.1 Differential Privacy

We implement DP-SGD [9] with:

- Per-example gradient clipping: $\|\nabla \ell(w; x)\|_2 \leq C$
- Gaussian noise addition: σ calibrated for (ϵ, δ) -DP
- Privacy budget: $\epsilon = 8$, $\delta = 10^{-5}$ (configurable)

3.3.2 Secure Aggregation

Using the protocol from Bonawitz et al. [10]:

- Clients encrypt updates using shared secret keys
- Server computes aggregate without decrypting individual updates
- Dropout resilience through threshold cryptography

4 Business Case and Adoption Strategy

4.1 Value Proposition

4.1.1 For Enterprises

Regulatory Compliance:

- **GDPR:** No source code transfer to third parties maintains data sovereignty
- **HIPAA:** Healthcare organizations can participate without exposing PHI in code
- **Financial Regulations:** Banks comply with data residency requirements (e.g., SOX, PCI DSS)
- **Government Contracts:** Meets ITAR, FedRAMP requirements for controlled data

Cost Reduction:

- Early vulnerability detection reduces remediation costs by 100x compared to post-deployment fixes [15]
- Reduced dependency on expensive penetration testing
- Lower insurance premiums through demonstrable security posture

Competitive Advantage:

- Collaborative learning improves model quality without data sharing
- Reduced time-to-market through faster security validation
- Enhanced reputation through proactive security measures

4.1.2 For Developers

- Immediate, contextual feedback reduces context switching
- Educational value: learning secure coding practices in real-time
- Reduced technical debt from security issues

4.2 Market Analysis

Total Addressable Market: The global application security market is projected to reach \$13.2 billion by 2027 (CAGR 21.3%) [16]. The federated learning market is growing at 8.3% CAGR, reaching \$320 million by 2028.

Target Segments:

1. Financial services (30% market share priority)
2. Healthcare technology (25%)
3. Government contractors (20%)
4. Enterprise SaaS providers (15%)
5. Other industries (10%)

4.3 Competitive Positioning

Table 1: Competitive Comparison

Feature	SecureCode-FL	Snyk	Veracode	SonarQube
Real-time Detection			×	
ML-Based		~	~	×
On-Premise Data		×	×	
Collaborative Learning		×	×	×
IDE Integration			~	
Continuous Adaptation		×	×	×

4.4 Pricing Strategy

Tiered Model:

- **Community Edition:** Free, single-user, basic vulnerability detection
- **Enterprise Edition:** \$50/developer/month, federated learning, advanced privacy features
- **Consortium Edition:** Custom pricing, dedicated federation server, white-labeling

5 Implementation Roadmap

5.1 Phase 1: Proof of Concept (Months 1-3)

Objectives:

- Develop baseline vulnerability detection model
- Implement basic VS Code extension
- Validate federated learning convergence

Deliverables:

1. Trained baseline model on public datasets (Juliet, SARD)
2. VS Code extension with local inference capability
3. Simulated federated learning environment with 10 nodes
4. Performance benchmarks: accuracy, latency, false positive rate

Technical Stack:

- ML Framework: PyTorch with FL libraries (Flower, PySyft)
- Extension: TypeScript, VS Code Extension API
- Model Server: Python FastAPI, gRPC for efficient communication
- Parsing: tree-sitter for multi-language AST generation

5.2 Phase 2: Alpha Development (Months 4-6)

Objectives:

- Integrate privacy-preserving mechanisms
- Expand language support (Python, JavaScript, Java initially)
- Optimize inference latency ($< 100\text{ms}$)

Deliverables:

1. Differential privacy implementation with configurable budgets
2. Secure aggregation protocol
3. Multi-language AST processing pipeline
4. Performance optimization: model quantization, TorchScript compilation

5.3 Phase 3: Beta Testing (Months 7-9)

Objectives:

- Conduct pilot deployments with 3-5 partner organizations
- Validate regulatory compliance claims
- Refine user experience based on feedback

Deliverables:

1. Beta program with real-world codebases
2. Compliance documentation (GDPR, HIPAA, SOC 2)
3. User feedback integration and UX improvements
4. Scalability testing: support for federations with 50+ participants

5.4 Phase 4: Production Release (Months 10-12)

Objectives:

- Launch production-ready extension on VS Code Marketplace
- Establish commercial federation infrastructure
- Implement monitoring and analytics platform

Deliverables:

1. Public release with comprehensive documentation
2. Enterprise deployment guides
3. Federation management dashboard
4. Telemetry system (privacy-preserving usage analytics)

6 Technical Challenges and Solutions

6.1 Challenge 1: Model Size and Inference Latency

Problem: Large neural networks cause IDE responsiveness issues.

Solutions:

- Model distillation: Train smaller student models from larger teachers
- Quantization: INT8 inference reducing model size by 4x
- Selective analysis: Focus on changed code regions
- Asynchronous processing: Non-blocking inference with caching

6.2 Challenge 2: Non-IID Data Distribution

Problem: Organizations have heterogeneous codebases, causing model bias.

Solutions:

- FedProx algorithm [17]: Add proximal term to local objectives
- Personalization layers: Organization-specific fine-tuning heads
- Adaptive aggregation: Weight clients by data quality metrics

6.3 Challenge 3: Cold Start Problem

Problem: New organizations lack labeled vulnerability data.

Solutions:

- Pre-trained foundation model on public datasets
- Active learning: Prioritize samples for expert labeling
- Weak supervision: Leverage existing static analysis tools for pseudo-labels
- Transfer learning: Adapt from similar organizations (with consent)

6.4 Challenge 4: Communication Efficiency

Problem: Frequent model updates consume bandwidth.

Solutions:

- Gradient compression: Top-k sparsification (communicate 1% of parameters)
- Federated dropout: Only update subset of model parameters
- Adaptive scheduling: Update frequency based on code change rate

6.5 Challenge 5: Byzantine Attacks

Problem: Malicious clients could poison the global model.

Solutions:

- Robust aggregation: Krum, Trimmed Mean algorithms
- Reputation systems: Weight trusted organizations higher
- Anomaly detection: Identify outlier updates before aggregation
- Verification: Test global model on curated validation set

7 Evaluation Plan

7.1 Performance Metrics

7.1.1 Detection Accuracy

- Precision, Recall, F1-score per vulnerability category
- False Positive Rate (target: $< 5\%$)
- True Positive Rate (target: $> 90\%$)

7.1.2 System Performance

- Inference latency (target: $< 100\text{ms}$ per file)
- Extension memory footprint (target: $< 200\text{MB}$)
- Model update bandwidth (target: $< 10\text{MB}$ per update)

7.1.3 Federated Learning Metrics

- Convergence rate: rounds to target accuracy
- Communication cost: total bytes transferred
- Privacy guarantees: achieved (ϵ, δ) values

7.2 Experimental Design

Datasets:

1. Juliet Test Suite: 64,000 synthetic C/C++ vulnerabilities
2. SARD: Software Assurance Reference Dataset
3. Real-world CVEs: Extracted from GitHub commits
4. Partner codebases: Proprietary (anonymized metrics only)

Baselines:

- Centralized learning (upper bound)
- Static analysis tools (SonarQube, CodeQL)
- Previous ML approaches (VulDeePecker, Devign)
- Local-only learning (lower bound)

Ablation Studies:

- Impact of differential privacy on accuracy
- Effect of federation size on model quality
- Contribution of different model components (GNN vs. Transformer)

8 Ethical Considerations

8.1 Privacy Implications

While federated learning preserves privacy by design, we must consider:

- **Model Inversion Attacks:** Adversaries might reconstruct training data from model parameters. Mitigation: Differential privacy, secure aggregation.
- **Membership Inference:** Determining if specific code was in training set. Mitigation: DP guarantees bound information leakage.
- **Transparency:** Organizations should understand what data contributes to learning. Mitigation: Explainable updates, audit logs.

8.2 Bias and Fairness

- **Dataset Bias:** Historical vulnerabilities may reflect biases in discovery processes. Mitigation: Diverse training data, fairness metrics.
- **Language Bias:** Over-representation of certain programming languages. Mitigation: Balanced multi-language training.
- **False Positives:** Disproportionate flagging of certain coding styles. Mitigation: Configurable thresholds, user feedback loops.

8.3 Responsible Disclosure

Organizations participating in federation might detect novel vulnerability patterns. Establishing responsible disclosure protocols is essential:

- Coordination with security research community
- Vendor notification timelines
- Public CVE disclosure after patches available

9 Future Work

9.1 Short-term Enhancements (6-12 months)

- **Multi-modal Analysis:** Incorporate code comments, documentation
- **Automated Repair:** Generate vulnerability patches using program synthesis
- **IDE Expansion:** Support for IntelliJ, PyCharm, Eclipse
- **Language Coverage:** Expand to C#, Go, Rust, Ruby

9.2 Long-term Vision (1-3 years)

- **Cross-organizational Transfer Learning:** Safely transfer knowledge between industry sectors
- **Adversarial Robustness:** Train models resilient to adversarial code obfuscation
- **Zero-shot Detection:** Identify novel vulnerability types without labeled examples
- **Blockchain Integration:** Decentralized, trustless federation coordination
- **Continuous Certification:** Automated security compliance verification

9.3 Research Directions

- **Theoretical Analysis:** Privacy-utility tradeoffs in code analysis
- **Optimization:** Communication-efficient aggregation for large models
- **Personalization:** Adapting global models to organization-specific contexts
- **Interpretability:** Explainable vulnerability predictions for developers

10 Conclusion

SecureCode-FL represents a significant advancement in automated vulnerability detection by combining the power of machine learning with the privacy guarantees of federated learning. The system addresses a critical market need: organizations require sophisticated security tools that respect data sovereignty and regulatory constraints. By enabling collaborative learning without data sharing, SecureCode-FL allows organizations to benefit from collective intelligence while maintaining complete control over their intellectual property.

The business case is compelling. Regulatory pressures (GDPR, HIPAA, financial regulations) create strong demand for privacy-preserving solutions. Early vulnerability detection provides measurable cost savings. The federated approach enables continuous model improvement, adapting to emerging threats without manual rule updates.

Our implementation roadmap demonstrates feasibility. The 12-month plan progresses from proof-of-concept through production deployment, with clear milestones and deliverables. Technical challenges—model size, heterogeneous data, communication efficiency—have established solutions from federated learning research. The VS Code extension architecture provides immediate value while enabling future expansion.

Looking forward, SecureCode-FL can evolve beyond vulnerability detection to comprehensive code quality analysis, automated repair, and continuous compliance verification. The federated learning paradigm opens new possibilities for privacy-preserving collaboration in software engineering.

In an era where software security and data privacy are paramount, SecureCode-FL offers a path forward: leveraging collective intelligence while respecting individual autonomy. This project has the potential to fundamentally change how organizations approach code security, making advanced AI-powered protection accessible without compromising on privacy.

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A Appendix A: Model Architecture Details

A.1 Graph Neural Network Component

The GNN processes code as a heterogeneous graph $G = (V, E)$ where nodes represent code elements (statements, expressions, variables) and edges represent dependencies (control flow, data flow, call graphs).

Node Feature Extraction:

$$x_v = \text{Embed}(\text{type}(v)) \oplus \text{Embed}(\text{token}(v)) \quad (4)$$

where $\oplus$ denotes concatenation.

Message Passing: For L layers:

$$h_v^{(l+1)} = \text{ReLU} \left(\sum_{t \in \mathcal{T}} \sum_{u \in \mathcal{N}_t(v)} W_t^{(l)} h_u^{(l)} + b^{(l)} \right) \quad (5)$$

where $\mathcal{T}$ represents edge types (control flow, data dependency, etc.) and $\mathcal{N}_t(v)$ are neighbors of type t .

Graph Readout:

$$h_G = \text{MeanPool}(\{h_v^{(L)} : v \in V\}) \oplus \text{MaxPool}(\{h_v^{(L)} : v \in V\}) \quad (6)$$

A.2 Transformer Encoder Component

Processes tokenized code sequences with positional encodings:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (7)$$

Multi-head Attention:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (8)$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$.

A.3 Classification Head

Multi-label vulnerability classification:

$$p(y|x) = \sigma(W_{\text{out}}[h_G \oplus h_{\text{seq}}] + b_{\text{out}}) \quad (9)$$

where σ is sigmoid activation for multi-label output.

Loss Function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C [y_{ij} \log p_{ij} + (1 - y_{ij}) \log(1 - p_{ij})] \quad (10)$$

with optional focal loss weighting for class imbalance:

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C \alpha_j (1 - p_{ij})^\gamma y_{ij} \log p_{ij} \quad (11)$$

B Appendix B: VS Code Extension Architecture

B.1 Extension Components

```
// Extension Structure
securecode-fl/
    client/                                // VS Code extension
        src/
            extension.ts                    // Entry point
```

```

        diagnostics.ts      // Vulnerability reporting
        codeActions.ts     // Quick fixes
        statusBar.ts       // UI elements
server/                    // Language server
    src/
        server.ts          // LSP implementation
        analyzer.ts        // Code analysis
        mlInference.ts     // Model inference
ml/                         // ML components
    model/
        gnn.py             // Graph neural network
        transformer.py     // Sequence encoder
        classifier.py      // Vulnerability classifier
    federated/
        client.py          // FL client logic
        server.py          // FL aggregation server
shared/                    // Shared utilities
    ast_parser.py         // AST extraction
    preprocessing.py      // Feature engineering

```

B.2 Communication Protocol

The extension uses gRPC for efficient client-server communication:

Inference Request:

```

message InferenceRequest {
    string file_path = 1;
    string code_content = 2;
    string language = 3;
    repeated string changed_lines = 4;
}

message InferenceResponse {
    repeated Vulnerability vulnerabilities = 1;
    float confidence = 2;
    int32 inference_time_ms = 3;
}

message Vulnerability {
    string type = 1;                // e.g., "SQL_INJECTION"
    int32 line_start = 2;
    int32 line_end = 3;
    float severity = 4;             // 0.0 to 1.0
    string description = 5;
    repeated string suggestions = 6;
}

```


C Appendix C: Federated Learning Protocol Specification

C.1 Communication Messages

Model Update Message:

```
message ModelUpdate {
    string client_id = 1;
    int32 round_number = 2;
    bytes model_weights = 3;          // Serialized tensors
    int32 num_samples = 4;
    float training_loss = 5;
    map<string, float> metrics = 6;
    bytes signature = 7;             // Cryptographic signature
}
```

Aggregation Response:

```
message GlobalModel {
    int32 round_number = 1;
    bytes model_weights = 2;
    float global_loss = 3;
    int32 num_participants = 4;
    map<string, float> global_metrics = 5;
}
```

C.2 Privacy Budget Tracking

Each client maintains a privacy accountant:

```
class PrivacyAccountant:
    def __init__(self, target_epsilon, target_delta):
        self.target_epsilon = target_epsilon
        self.target_delta = target_delta
        self.spent_epsilon = 0.0

    def compute_epsilon(self, noise_multiplier,
                       num_steps, sampling_rate):
        # Moments accountant or RDP accountant
        epsilon = compute_rdp_epsilon(
            noise_multiplier, num_steps,
            sampling_rate, self.target_delta
        )
        return epsilon

    def can_train(self):
        return self.spent_epsilon < self.target_epsilon
```

D Appendix D: Deployment Guide

D.1 System Requirements

Client Side (Developer Workstation):

- VS Code 1.80+
- Python 3.8+ (for local model server)
- 4GB RAM (8GB recommended)
- 2GB free disk space
- Network: 100 Kbps minimum for federated updates

Federation Server:

- Linux server (Ubuntu 20.04+ or RHEL 8+)
- 16+ CPU cores
- 64GB+ RAM
- 1TB SSD storage
- 1Gbps network connection
- NVIDIA GPU (optional, for faster aggregation testing)

D.2 Installation Steps

Client Installation:

```
# Install VS Code extension
code --install-extension securecode-fl-0.1.0.vsix

# Install Python dependencies
pip install securecode-fl-client

# Initialize local configuration
securecode-fl init --federation-server https://fl.example.com
```

Server Deployment:

```
# Using Docker
docker pull securecode-fl/server:latest
docker run -d \
  -p 8080:8080 \
  -p 50051:50051 \
  -v /data/fl:/data \
  --name securecode-fl-server \
  securecode-fl/server:latest

# Or using Kubernetes
kubectl apply -f k8s/securecode-fl-deployment.yaml
```

D.3 Configuration

Client Configuration (‘.securecode-fl/config.yaml’):

```
federation:
  server_url: "https://fl.example.com"
  update_interval: 3600 # seconds

privacy:
  enable_differential_privacy: true
  epsilon: 8.0
  delta: 1e-5

inference:
  mode: "real-time" # or "on-demand"
  languages: ["python", "javascript", "java"]
  min_confidence: 0.7

training:
  enable_local_training: true
  batch_size: 32
  local_epochs: 3
  learning_rate: 0.001
```

D.4 Monitoring and Maintenance

Key Metrics to Monitor:

- Federation health: active clients, update success rate
- Model performance: accuracy, F1-score per vulnerability type
- System performance: inference latency, memory usage
- Privacy metrics: epsilon consumption rate

Maintenance Tasks:

- Weekly: Review false positive reports, update validation set
- Monthly: Re-evaluate privacy budgets, perform security audits
- Quarterly: Major model updates, add new vulnerability patterns

E Appendix E: Sample Vulnerability Detection Output

Example: SQL Injection Vulnerability Detection

```
def get_user_data(username):
    # WARNING: SQL Injection vulnerability detected
    # Severity: HIGH (0.92 confidence)
```

```

# Line 4: Unsanitized user input in SQL query
query = f"SELECT * FROM users WHERE name = '{username}'"
return db.execute(query)

# Suggested fix:
# Use parameterized queries:
# query = "SELECT * FROM users WHERE name = ?"
# return db.execute(query, (username,))

```

Detection Output in VS Code:

[SecureCode-FL] 1 vulnerability found:

```

Line 4: SQL Injection
Severity: HIGH
Confidence: 92%
CWE: CWE-89
Suggestion: Use parameterized queries
Code Action: Apply automatic fix

```